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Vacuum evaporative heat exchanger

Abstract

Techniques are described herein for using a high-pressure reactor to separate clean water from dirty water without filtration and to extract and concentrate contaminants from dirty water for use as a fuel. In particular, techniques and systems are described for separating water from hydrocarbon contaminates, other BTU-laden compounds, and dissolved minerals, while also boiling water and condensing the resulting steam into distilled water. In addition, system in which the described techniques are performed can be used as a high-pressure pump for moving the separated hydrocarbon contaminates forward into other processes, such as a high-pressure reactor or incinerator.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 17/651,342, filed Feb. 16, 2022, and titled, "VACUUM EVAPORATIVE HEAT EXCHANGER," which claims the benefit of and priority to U.S. Provisional Application No. 63/200,125, filed Feb. 16, 2021, and titled, "VACUUM EVAPORATIVE HEAT EXCHANGER," these applications are incorporated by reference herein in their entireties.

TECHNICAL FIELD

(1) This disclosure relates to techniques for purifying water and, in particular, to purification of water with a vacuum evaporative heat exchanger.

BACKGROUND

(2) Access to clean water, and the removal of contaminants from water to create clean water, is an increasing need in the world. Many common water contaminants include hydrocarbons, which maybe provide a source of energy if they are removed from the water. Examples of contaminated water include oil production water, paper pulp slurry, food and rendering plant wash down, biofuel process waste fluids, sewer water, animal waste wash down water, and dairy wash down water. If the water can be separated from the contaminants, then the water can be reused, and the contaminants may be used as fuel.

SUMMARY

(3) Techniques are described herein for using a high-pressure reactor to separate clean water from dirty water without filtration and to extract and concentrate contaminants from dirty water for use as a fuel. In particular, techniques and systems are described for separating water from hydrocarbon contaminants, other BTU-laden compounds, and dissolved minerals, while also boiling water and condensing the resulting steam into distilled water. In addition, system in which the described techniques are performed can be used as a high-pressure pump for moving the separated hydrocarbon contaminants forward into other processes, such as a high-pressure reactor or incinerator.

(4) In some aspects, the techniques described herein relate to a system including: an oxidation reactor configured to oxidize combustible fuel to produce heat and hot steam as a result of the oxidation of the fuel; a first chamber having a hollow first cavity configured to be loaded with water and the fuel; a vacuum pump operationally connected to the hollow first cavity and configured to draw a vacuum on the hollow first cavity to lower a boiling point of water in the hollow first cavity to a lowered boiling point at least 20° F. below the boiling point of water at atmospheric pressure; a second chamber including a hollow second cavity that at least partially surrounds the hollow first cavity, the hollow second cavity being operationally connected to the oxidation reactor to receive the hot steam from the oxidation reactor, such that the hot steam in the hollow second cavity causes water in the hollow first cavity to boil to cold steam at the lowered boiling point; a movable piston located within the hollow first cavity and in scalable contact with an inner wall of the first chamber and configured to move along a length of the first chamber to mechanically move residual fuel in the hollow first cavity to the oxidation reactor, wherein the residual fuel remains in the first cavity after water in the hollow first cavity boils to cold steam before it is moved to the oxidation reactor; and a hydraulic pump operationally connected to the hollow first cavity and configured to pump liquid in and out of the hollow first cavity to move the movable piston between a first position in which the fuel and water is loaded into the hollow first cavity and a second position in which the residual fuel is moved from the hollow first cavity to the oxidation reactor.

(5) In some aspects, the system further includes a heat exchanger configured for receiving the cold steam from the hollow first cavity and condensing the cold steam into a liquid.

(6) In some aspects, the system further includes a steam condensate tank operationally connected to the hollow second cavity and configured for receiving condensate produced in response to heat transferred from the hot steam in the hollow second cavity to the water and fuel in the hollow first cavity.

(7) In some aspects, the system further includes a drive system operationally coupled to the oxidation reactor and to the hollow second cavity between the oxidation reactor and the hollow second cavity and configured for receiving the hot steam from the oxidation reactor and converting energy in the hot steam into mechanical energy and then passing the hot steam to the hollow second cavity.

(8) In some aspects, the oxidation reactor is configured to produce heat and hot steam as a result of the oxidation of the fuel at a temperature of greater than 500° F.

(9) In some aspects, the lowered boiling point is at least 110° F. below the boiling point of water at atmospheric pressure.

(10) In some aspects, the hollow first cavity includes a first end operationally coupled to the hydraulic pump and a second end operationally coupled to the vacuum pump, the first end and the second end being separated by the movable piston in the hollow first cavity.

(11) In some aspects, the movable piston includes stainless steel and a sealing surface on an outer surface of the piston that seals against an inner wall of the hollow first cavity, wherein the sealing surface is configured to remove condensate material from the inner wall of the hollow first cavity.

(12) In some aspects, the first chamber includes at least one copper wall defining the hollow first cavity.

(13) In some aspects, the second chamber includes at least one copper wall defining the hollow second cavity.

(14) In some aspects, the techniques described herein relate to a method including: oxidizing combustible fuel in an oxidation reactor to produce heat and hot steam as a result of the oxidation of the fuel; loading a hollow first cavity of a first chamber that is operationally connected to the oxidation reactor with water and the fuel; drawing a vacuum on the hollow first cavity to lower a boiling point of water in the hollow first cavity to a lowered boiling point at least 20° F. below the boiling point of water at atmospheric pressure; providing the hot steam from the oxidation reactor to a hollow second cavity of a second chamber that at least partially surrounds the hollow first cavity, such that the hot steam in the hollow second cavity causes water in the hollow first cavity to boil to cold steam at the lowered boiling point; moving a movable piston located within the hollow first cavity and that is in scalable contact with an inner wall of the first chamber along a length of the first chamber to mechanically move residual fuel in the hollow first cavity to the oxidation reactor, wherein the residual fuel remains in the first cavity after water in the hollow first cavity boils to cold steam before it is moved to the oxidation reactor, wherein the residual fuel that is moved into the oxidation reactor provides the combustible fuel for oxidation in the oxidation reactor; and pumping liquid in and out of the hollow first cavity to cause the movement of the movable piston between a first position in which the fuel and water is loaded into the hollow first cavity and a second position in which the residual fuel is moved from the hollow first cavity to the oxidation reactor.

(15) In some aspects, the method further includes condensing the steam from the hollow first cavity into a liquid.

(16) In some aspects, the method further includes receiving into a steam condensate tank operationally connected to the hollow second cavity condensate produced in response to heat transferred from the hot steam in the hollow second cavity to the water and fuel in the hollow first cavity.

(17) In some aspects, the method further includes: receiving into a drive system operationally coupled to the oxidation reactor and to the hollow second cavity between the oxidation reactor and the hollow second cavity the hot steam from the oxidation reactor; converting energy in the hot steam into mechanical energy; and passing the hot steam to the hollow second cavity.

(18) In some aspects, the oxidation of the fuel in the oxidation reactor occurs at a temperature of greater than 500° F.

(19) In some aspects, the lowered boiling point is at least 110° F. below the boiling point of water at atmospheric pressure.

(20) In some aspects, the hollow first cavity includes a first end operationally coupled to the hydraulic pump and a second end operationally coupled to the vacuum pump, the first end and the second end being separated by the movable piston in the hollow first cavity.

(21) In some aspects, the movable piston includes stainless steel and a sealing surface on an outer surface of the piston that seals against an inner wall of the hollow first cavity, wherein the sealing surface is configured to remove condensate material from the inner wall of the hollow first cavity.

(22) In some aspects, the first chamber includes at least one copper wall defining the hollow first cavity.

(23) In some aspects, the second chamber includes at least one copper wall defining the hollow second cavity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of a system for creating clean water from contaminated water.
- (2) FIG. 2A is a schematic diagram of an example first chamber defined by a wall having a circular cross section contained within an example second chamber defined by a wall with circular cross section.
- (3) FIG. 2B is a schematic diagram of multiple example first chambers defined by walls having a circular cross section contained within an example second chamber defined by a wall with circular cross section.
- (4) FIG. 2C is a schematic diagram of multiple example first chambers defined by walls having a circular cross section contained within an example second chamber defined by a wall with a rectilinear cross section.
- (5) FIG. 3 is a flow chart of a process for purifying contaminated water.

DETAILED DESCRIPTION

- (6) FIG. 1 is a schematic diagram of a system **100** for creating clean water from contaminated water. The system **100** includes, among other things, a storage tank **11** for storing contaminated water, a hollow chamber **1** into which contaminated water is injected and in which fuel is separated from water, a high-pressure, high-temperature oxidation reactor **14** that consumes fuel extracted from the contaminated water and that produces heat, and a second chamber **2** that surrounds the first chamber **1** forming a jacket around the first chamber **1** and to which hot steam from the high-temperature reactor can be supplied. In some implementations, the high-pressure, high-temperature oxidation reactor **14** can include a super-critical water oxidation reactor. In some implementations, the high-pressure, high-temperature oxidation reactor **14** can burn fuel that is extracted from the contaminated water.
- (7) The second chamber **2** can at least partially surround the first chamber **1**, such that the second chamber **2** forms a jacket around the first chamber **1**. In some implementations, the first chamber **1** can include a first cylindrical tube and the second chamber **2** can include a second cylindrical tube, and the tubes can be concentric along a common axis. The first chamber **1** and the second chamber **2** each can have an inner wall diameter and an outer wall diameter, with the inner wall diameter of the second chamber **2** being larger than the outer wall diameter of the first chamber, such that a cavity is formed between the outer wall of the first chamber **1** and the inner wall of the second chamber **2**. In an implementation, the first chamber **1** and the second chamber **2** can be straight, but the tubes need not be straight. The first and second chambers need not have circular cross sections but can have other shapes as well, including rectilinear cross sections. FIG. 2A is a schematic diagram of an example first chamber defined by a wall **200** having a circular cross section contained within an example second chamber defined by a wall **202** with circular cross section.
- (8) The first chamber **1** includes a sliding plunger (e.g., floating piston) **3** that includes a wiper seal that seals against an inner wall of the first chamber **1**. Contaminated liquid is loaded into the first chamber, and a vacuum pump **18** evacuates gas from within the chamber **1** to reduce a pressure in the chamber **1**, thus pulling a vacuum on the contents of the chamber **1** and reducing the boiling point of the liquid in the first chamber **1**. Steam is passed into the second chamber **2** in the cavity between the outer wall of the first chamber **1** and the inner wall of the second chamber **2**, and the steam condenses on the outer wall of the first chamber **1**. The heat from the steam in the second chamber **2** and the vacuum pressure in the first chamber causes the liquid in the first chamber **1** to boil, leaving behind the concentrates and minerals that were dissolved in the liquid. These

concentrates are then pushed out of the first chamber **1** and into an oxidation reactor **14** or in another system for further processing. Steam from the boiled liquid from within the first chamber **1** can be collected and condensed, thus separating, and purifying, the liquid that was injected into the chamber from the contaminants in liquid.

(9) Walls of the first chamber **1** can be made of, for example, copper, stainless steel, glass, ceramic, carbon fiber, or other appropriate material. Inner walls of the chamber **1** can be lined with a non-reactive material, such as, for example, ceramic or PTFE.

(10) The second chamber **2** (e.g., an outer tube or jacket) can surround the first chamber **1** for the purpose of containing steam in a cavity between an inner wall of the second chamber in an outer wall of the first chamber, such that the steam can condense against the inner chamber **1**. Metal fins or metal mesh can be attached to the outer wall of the first chamber and used to enhance the heat transfer from the steam in the second chamber **2** to contents within the first chamber **1**.

(11) A piston **3** can slide within the first chamber **1**, making a contact seal with the inner wall of the first chamber. The piston **3** can be made of copper, stainless steel, glass, ceramic, carbon fiber, or other appropriate material, and can include one or more surfaces that make a sealing contact with the inner wall of the second chamber **2**. The sealing surfaces can include, for example, PTFE or some other suitable material that can withstand higher temperatures without melting and sticking to the inner wall of the first chamber **1**. In some implementations, the piston **3** can include a stainless steel cylinder (or other shape that matches the cross section of the inner wall of the second chamber) and can have one or more o-ring sealing surfaces on its outer surface that make a sealing contact with the inner wall of the second chamber **2** to create a water- and air-tight seal between portions of the first chamber **1** on opposite sides of the piston **3**.

(12) The second chamber **2** includes a discharge point **4** through which condensate (e.g., water) from the steam that condenses within the second chamber **2** can exit the chamber **2**.

(13) A first end portion of the first chamber **1** can include a reducer **5** (e.g., portion of the first chamber having an inner diameter that is smaller than an inner diameter of a main portion of the chamber) through which contaminated liquid flows into a main portion of the first chamber **1** and through which concentrated fuel flows out of a main portion of the first chamber **1**. The reducer **5** can prevent the piston **3** from traveling beyond a first end and leaving the main portion of the first chamber **1**, because its diameter is smaller than that of the piston **3**.

(14) An inlet port **6** on second chamber **2** can receive hot steam emitted from the oxidation reactor **14** into the second chamber **2**, in some cases, after energy of the hot steam has been used to perform work in a mechanical drive system **15**.

(15) A second end portion of the first chamber **1** can include a reducer **7** (e.g., a portion of the chamber **1** having an inner diameter that is smaller than an inner diameter of a main portion of the chamber) on the piston hydraulic drive side which prevents the piston **3** from traveling beyond a second end of chamber **1**.

(16) A control valve **8** between the first chamber **1** and the contaminated supply water tank/hopper **11** can be operated to allow contaminated liquid to pass from a storage tank or hopper **11** that stores contaminated water into the first chamber **1** when the valve **8** is opened. When the control valve **8** is closed, it can allow vacuum to be drawn on the first chamber **1** and can prevent concentrates from being pushed back to the contaminated supply water tank/hopper **11**.

(17) A control valve **9** between the first chamber **1** and the oxidation reactor **14** can be operated to allow vacuum to be drawn on chamber **1** and to prevent high-pressure reactor fluid from traveling backwards from the reactor into chamber **1** when the valve **9** is closed. When the valve **9** is open, it can allow pressurized concentrates to move from the first chamber **1** into the high-pressure oxidation reactor **14**.

(18) A pump **10** operably connected to the contaminated supply water tank/hopper **11** can be operated to move the contaminated water from the tank/hopper **11** through valve **8** and into the first chamber **1**.

- (19) A control valve **12** operably connected to the high-pressure oxidation reactor **14**, when closed, forces high-pressure concentrate that passes through valve **9** from the first chamber **1** into the high-pressure oxidation reactor **14**. When the valve **12** is open, it can allow combustible fuel and other materials from additional systems connected beyond a connection point **13** to be pumped into the high-pressure oxidation reactor **14**.
- (20) The high-pressure, high-temperature oxidation reactor **14** can combust, oxidize, or otherwise react the concentrated hydrocarbons and other minerals that are received from chamber **1** into the oxidation reactor **14** to produce heat, clean water, and fully broken-down hydrocarbon molecules and metal oxides. The oxidation reactor **14** can operate at pressures of hundreds to thousands of PSI and can operate at temperatures of higher than 700 degrees Fahrenheit.
- (21) In some implementations, a mechanical drive system **15** operably connected to the high-pressure, high-temperature oxidation reactor **14** can use energy in hot gas, supercritical water, or high temperature steam (e.g., from oxidation reactor **14**) to perform mechanical work, such as to generate electricity, drive compressors, create motion, etc.
- (22) A control valve **16** between the second chamber **2** and the oxidation reactor **14** can be operated to allow energy (e.g., steam) from oxidation reactor **14** or from the drive system **15** to pass into the chamber **2**.
- (23) In some implementations, the system **100** can include multiple first chambers **1** that are contained within one or more second chambers **2**. FIG. 2B is a schematic diagram of multiple example first chambers defined by walls **210** having a circular cross section contained within an example second chamber defined by a wall **212** with circular cross section. FIG. 2C is a schematic diagram of multiple example first chambers defined by walls **220** having a circular cross section contained within an example second chamber defined by a wall **222** with a rectilinear cross section. When running multiple systems that include multiple pairs of first chambers **1** and/or second chambers **2**, the system **100** can include multiple valves **16** that can be used to determine which of the chambers the steam is directed to.
- (24) A control valve **17** between the first chamber **1** and condensate heat exchanger **20** is positioned such that when closed it blocks the flow of high-pressure concentrate from moving from the first chamber **1** into a heat exchanger **20**. When open, the control valve **17** allows the vacuum steam from first chamber **1** to travel to the heat exchanger **20**. The condensate heat exchanger **20** can receive vacuum generated steam from the first chamber **1** and then cool the steam to condensate using cooler water or glycol that enters the heat exchanger **20** from port **38** and leaves the heat exchanger **20** from port **39**. Ports **38**, **39** in the heat exchanger **20** allow cooling fluid to pass through the heat exchanger **20**. In some implementations, the system **100** can include a catchment hopper **43** located between the valve **17** and heat exchanger **20**, which can catch boil-over material from the chamber **1**. For example, the hopper **43** can receive liquid and gas from the chamber and can include an output port at the top of the hopper through which gas (e.g., vacuum steam) is provided to the heat exchanger. Liquid can be drained from the hopper **43** through drain port that is opened by valve **44**.
- (25) A vacuum pump **18** can be operationally connected to first chamber **1** and can draw down the pressure in chamber **1** to a pressure less than atmospheric pressure (e.g., by more than 500 torr below atmospheric pressure). In some implementations the pump **18** can be connected to the first chamber **1** through a condensate tank **22** and the heat exchanger **20**. A high vacuum check valve **19** between the pump **18** and the first chamber **1** can be closed to maintain vacuum on the chamber **1** and the heat exchanger **20** even when pump **18** is turned off. The condensate tank **22** can be used for collecting the vacuum distilled water that is created in the first chamber **1** and condensed in the heat exchanger **20**. The tank **22** can be created and configured to hold a vacuum pressure when gas is pumped out of it by pump **18**.
- (26) A control valve **21** located between the heat exchanger **20** and the condensate tank **22** can be controlled to allow the condensate from the exchanger **20** to drain into the vacuum condensate tank

22. When closed, the valve **21** allows vacuum to be maintained on chamber **1** and heat exchanger **20** while condensate tank **22** is drained. A port connection **23** connects the tank to atmospheric air, and air that flows into the tank **22** can be filtered before entering tank **22** through valve **24**. The control valve **24** can allow atmospheric air into the condensate tank **22**, so that the tank can be drained of condensate, and when the valve **24** is closed atmospheric air is blocked from the condensate tank allowing vacuum to build up in the condensate tank **22**.

(27) De-contaminated condensate (e.g., purified water) can be pumped out of the tank **22** by a pump **26** into a storage tank maintained at normal atmospheric pressure. A control valve **25** between the condensate tank **22** and the pump **26** can be closed to allow vacuum to build up in the tank.

(28) A steam condensate tank **27** can collect the steam condensate from the second chamber **2**. The tank **27** can be maintained at atmospheric pressure. A control valve **28** on an inlet port **31** to the tank **27** can be used to close off the atmospheric vent for condensate tank **27**. The port **31** allow air into tank **27**, where the air may pass through a filter (e.g., a HEPA rated filter) to prevent dust or contaminates from entering tank **27**. Air from the inlet port **31** allows the tank to remain at atmospheric pressure when valve **28** is open. The valve **28** is normally open to allow water to enter the tank and air to leave the tank or vice versa when the tank is being emptied through valve **30**. A pump **29** that acts as a condensate transfer pump can send the “hot steam” purified condensate forward to storage tanks or for use as process, irrigation, pre-drinking water, etc. A valve **30** between the tank **27** and the pump **29** is normally closed to contain the condensate in condensate tank **27**, and when open allows condensate to reach the pump **29**.

(29) A hydraulic pump **33** can be coupled to the first chamber **1** to increase the pressure in the chamber **1**. In some implementations, the pump **33** is capable of reaching pressures higher than the operating pressures inside oxidation reactor **14**, so that material from the chamber **1** can be pressurized to a pressure greater than that of the reactor for injection into the reactor. A control valve **32** that when open allows fluid from a hydraulic reservoir **34** containing a hydraulic fluid or water for use as a hydraulic fluid to reach pump **33**. The hydraulic pump **33** pushes fluid against the sliding piston **3** to force the piston through the first chamber **1** to move concentrate forward to oxidation reactor **14**, or the pump **33** draws hydraulic fluid back from the first chamber **1** to pull the piston **3** back to its starting place close to the reducer **7**. The pump **33** can be a fixed displacement or piston type pump.

(30) A control valve **35** allows hydraulic fluid to move from the pump **33** in and out of the first chamber **1** and to control the position of piston **3**. When closed, the valve **35** prevents the flow of hydraulic fluid in this loop. A control valve **36** allows hydraulic fluid being pumped by the pump **33** to flow into the reservoir **34**. A control valve **37** that when open allows the pump **33** to draw hydraulic fluid out of the first chamber **1** and to pull the piston **3** back toward the reducer **7**. When closed, the valve **37** allows fluid to build up pressure as it passes through control valve **35** to push the piston **3** back toward position **5**.

(31) In some implementations, the operating pressure of the first chamber **1** can be below that of the oxidation reactor **14**. For example, if the walls of the chamber **1** are made of soft, malleable materials (e.g., copper), operating a reduced pressure as a compared with that of the oxidation reactor **14** may be advantageous to prevent deformation of the shape of the chamber **1**. In some implementations, material expelled from the chamber **1** and passed through valve **9** can be collected in a chamber **40**, while control valve **41** is closed. Then, valve **9** can be closed, and pump **42** can pressurize the contents of the chamber **40** to a pressure close to, or exceeding, the operating pressure of the oxidation reactor **14**, and then injected into the reactor upon opening the valve **41**.

(32) The system **100** can be used for separating contaminates from dirty water, using the separated contaminates as fuel in a oxidation reactor **14**, and using the oxidation reactor **14** to produce power. Steam produced from the contaminated water, produced by the oxidation reactor **14** and by distilling the water, can be condensed in two streams of distilled and condensate water for use.

(33) Referring to FIG. 1, water containing contaminants can be deposited and stored in the tank or hopper **11**, and then pumped from the tank **11** through pump **10** and through control valve **8** into the first chamber **1**. The flow of the contaminated liquid from tank **11** can be blocked from going to elsewhere in the system **100** by the closure of valve **9**, the closure of valve **17**, and by opening valve **37**, while activating pump **33** and opening valve **36**. The pressure on the liquid from pump **10** and the vacuum on the liquid from pump **33** draws the piston **3** back along first chamber **1** to position **7**.

(34) A timer, or a fixed displacement metering system, connected to a programmable system controller **45**, can ensure that a predetermined amount of liquid from tank **11** passing through pump **10** does not completely fill first chamber **1** and that air space is left in the first chamber **1** to allow steam to pass across the top of the liquid from hopper **11**. Once the first chamber **1** is loaded with a measured amount of contaminated liquid, the pump **10** is shut off by the system controller **45**. Valve **8** is closed, while valves **9**, **17** remain closed. Valve **37** closes and valve **32** opens, and valve **36** opens, and pump **33** is shut down, so that piston **3** can move to the left in the first chamber **1** toward reducer **7**.

(35) Then, with valve **17** open and valves **24** and **25** closed, vacuum pump **18** is activated, and valve **21** is opened to draw vacuum over the contaminated water inside of the first chamber **1**. Activating vacuum pump **18** begins to draw air and gas from the top of the liquid in the first chamber **1**, and the gas and vapors are drawn through heat exchanger **20**. The vacuum created by pump **18** also creates a vacuum in the condensate collection tank **22**. This vacuum causes the boiling point of the liquid in the first chamber **1** to drop substantially, for example, to below 100 degrees Fahrenheit. When the system controller **45** has determined that the vacuum has reached a high enough level, then valve **16** is opened, allowing steam from the steam drive system **15** to flow through the second chamber **2**. Then, valve **28** can be opened to allow cooler air to flow into the second chamber **2** and to prevent back pressure on the steam entering the second chamber **2**.

(36) When steam is received from the drive system **15** into the second chamber **2**, such that the steam in the second chamber **2** passes around first chamber **1**, the steam is cooled by the cooler temperature of the wall of the first chamber (due, at least in part, to the evaporating liquid in the first chamber **1**) below the condensation point of the steam. This causes the steam in the second chamber **2** to turn into water as heat from the steam is transferred to the liquid inside of first chamber **1**.

(37) The liquid inside of the first chamber **1** begins to boil, as a result of the low pressure inside the chamber **1** and as a result of the heat supplied by the steam in the second chamber **2**. As the water in the first chamber **1** boils, the resulting steam travels along the surface of the liquid in first chamber **1**, and then up and through valve **17** and into the heat exchanger **20**. As the liquid is boiling in first chamber **1**, it leaves behind minerals and hydrocarbons, which began to concentrate in the first chamber **1**. As the steam reaches the heat exchanger **20** (e.g., at a temperature of approximately 80 to 120 degrees Fahrenheit, e.g, more than 110° F. below the boiling point of water at atmospheric pressure), a cooler liquid that passes through the heat exchanger **20** causes that steam to re-condense into liquid as it gives up heat to the heat exchanger **20**. As the steam becomes liquid again in the heat exchanger **20**, it creates additional vacuum, which continues to pull on the liquid in first chamber **1**. When the system controller **45** has determined that the condensate vacuum is high enough, vacuum pump **18** may be turned off, for the duration of the cycle, or may be turned on and off in order to maintain the correct level of vacuum on heat exchanger **20** and first chamber **1**. The liquid that condensates in heat exchanger **20** then travels down through valve **21** and into the condensate collection tank **22**, where it is allowed to build up until some predetermined level is reached.

(38) At that point, valve **25** may be opened, and pump **26** may be used to pump out the condensate. If, due to the vacuum pressure, the condensate pump **26** cannot overcome the negative pressure and it cavitates, then valve **24** can be opened after valve **21** is closed to allow the liquid to be pumped

out. Once a low level is reached in tank **22**, then valve **24** can be closed and valve **25** can be closed, and the pump **29** can be turned off.

(39) Once the condensate tank **22** has been drained, valve **21** can be reopened, vacuum pump **18** can be re-engaged, and vacuum can be re-established inside of tank **22** as well as through heat exchanger **20** and back to the first chamber **1**.

(40) When the system controller, using level sensors in tank **22**, and/or temperature sensors in the first chamber **1**, determines sufficient water has been boiled out of the concentrate in chamber **1**, the system can be changed to a discharge mode. To accomplish this, valve **17** is closed, valve **35** is opened, pump **33** is activated as valve **32** is opened. This forces hydraulic fluid from reservoir **34** to pass valve **35** and into the first chamber **1**, which pushes the piston **3** against the concentrate that has been left behind in the first chamber **1**. Valve **9** is then opened to allow the concentrate to be pushed by piston **3** into oxidation reactor **14**, using the pressure from pump **33**.

(41) The system controller **45** can control pump **33** through a variable frequency drive or other mechanism to control the rate at which the fluid passes through valve **35** and into the first chamber **1** and therefore the flow rate controls the rate at which the concentrate in the first chamber **1** is pushed into the oxidation reactor **14**. As the oxidation reactor **14** receives the fuel and concentrated minerals, it oxidizes these incoming components to produce high temperature water under high-pressure and significant heat. That heat energy is used by releasing super critical water, steam, or hot high-pressure gas, from oxidation reactor **14** to a drive system **15**. The drive system **15** may include a traditional steam piston drive arrangement, a turbine, or other super-critical water drive system. The drive system **15** may include some other mechanism that uses expanding supercritical water or steam to do work such as compression of gas or other expansion. When the steam has performed work in the drive system **15**, it is vented through valve **16** and into the second chamber **2** through inlet port **6**. The steam in the second chamber **2** condenses on the inner tube **3** and produces the condensate which is collected in condensate tank **27**.

(42) Typically valve **28** remains open, and when condensate tank **27** reaches a predetermined level, valve **30** is opened and pump **29** pumps this condensate out to either a storage tank or for use.

(43) The liquid in both tank **27** and tank **22** is first-pass distilled water, meaning it is very clean. The liquid in tank **27** is both clean and also completely sterile. The liquid in tank **22** is very clean, although may not be considered sterile without testing.

(44) When the piston **3** has reached the right side of the first chamber **1** near reducer **5**, the system is ready to be reset and loaded with new fuel and water for separating. At this point, valve **16** is closed, and steam from the drive **15** is directed to a different system or vented. Valve **9** is closed, valve **37** is open, valve **8** is open, and valve **36** is open, pump **33** is activated, and pump **10** is activated, allowing more of the liquid from tank **11** to be moved into first chamber **1**, for processing. In this process, again the piston **3** will move from a position near reducer **5** to a position near reducer **7**. And then the process will be restarted.

(45) While it may be possible to run the system on a single first chamber **1**, as shown in FIG. **1**, multiple sets of vacuum chambers **1** also can be used (with each chamber being surrounded by one or more second chambers **2**), for example, in a carousel design, so that fuel can be continuously delivered to oxidation reactor **14**, and so that steam from drive **15** can be continuously discharged through this system.

(46) Water, while undergoing a phase change, requires the transfer of heat to change temperature and phase. This is different or independent of temperature, although a temperature differential is required for the heat to transfer from one body to another. As described herein, the ability to place vacuum on a cooler liquid (e.g., water) and concentrate for separation, in the first chamber **1**, lowers the boiling point of the water in chamber **1** significantly. The spent steam coming from drive **15** carries with it not only a higher temperature (e.g., between 200 to 230 degrees Fahrenheit), it also carries with it latent heat, where the latent heat must be extracted from the steam for the steam to condense back to liquid. In general, the steam will not drop in temperature below its

boiling/condensation point until it has given up the latent heat that it carries, at which point it will cross the phase transition and become liquid. In other words, the temperature of the steam, at that transition point will remain constant until it has given up enough heat that the steam becomes liquid.

(47) In the first chamber **1**, the dirty water that is to be separated into clean water and concentrate, undergoes a similar phenomenon in reverse. The temperature of the dirty water rises until it reaches its boiling point, at which point its temperature will not rise any further until it has absorbed additional heat which allows the liquid water to boil to steam.

(48) By placing a vacuum on the first chamber **1**, the boiling point of the water in the solution is lowered from typically 212 degrees Fahrenheit at sea level or 204 degrees Fahrenheit at mountain elevations, to somewhere around 70-80 degrees Fahrenheit at 135 torr. Thus, the boiling point of the water in the solution can be lowered, in some implementations, to more than 110° F. below the boiling point of water at local atmospheric pressure.

(49) Thus, when the steam coming into the second chamber **2** with a temperature of over 200 degrees Fahrenheit contacts the outer wall of the first chamber **1** (or fins connected with the first chamber), which contains water that cannot rise above 70 or 80 degrees Fahrenheit because of the reduced boiling point caused by vacuum, heat is transferred from the steam in the outer jacket of the second chamber to the liquid in the first chamber **1**, causing the liquid in the first chamber **1** to boil at 70-80 degrees Fahrenheit. This heat transfer is very efficient because the temperature difference between the two fluids is so high. At around a temperature difference between the two liquids of about 100 degrees Fahrenheit, the heat will readily flow to the cooler liquid, thereby boiling the liquid and at the same time extracting enough heat from the steam in the second chamber **2** to condense the steam back into water; then the condensate can further lose both heat and temperature until its temperature approaches 90 to 100 F.

(50) The system controller **45** can include one or more memory devices storing computer readable instructions and one or more processors configured for executing the instructions. The instructions may be executed to programmatically control operation of the system **100**. For example, the timing of the opening and closing of the valves of the system **100** can be controlled according to the techniques described herein.

(51) FIG. **3** is a flow chart of a process **300** for purifying contaminated water. The process includes oxidizing combustible fuel in an oxidation reactor to produce heat and hot steam as a result of the oxidation of the fuel (**302**). The process includes loading a hollow first cavity of a first chamber that is operationally connected to the oxidation reactor with water and the combustible fuel (**304**). The process includes drawing a vacuum on the hollow first cavity to lower a boiling point of water in the hollow first cavity to a lowered boiling point at least 20° F. below the boiling point of water at atmospheric pressure (**306**). The process includes providing the hot steam from the oxidation reactor to a hollow second cavity of a second chamber that at least partially surrounds the hollow first cavity, such that the hot steam in the hollow second cavity causes water in the hollow first cavity to boil to cold steam at the lowered boiling point (**308**). The process includes moving a movable piston located within the hollow first cavity and that is in sealable contact with an inner wall of the first chamber along a length of the first chamber to mechanically move residual fuel in the hollow first cavity to the oxidation reactor, wherein the residual fuel remains in the first cavity after water in the hollow first cavity boils to cold steam before the residual fuel is moved to the oxidation reactor, wherein the residual fuel that is moved into the oxidation reactor provides the combustible fuel for oxidation in the oxidation reactor (**310**).

(52) The process includes pumping liquid in and out of the hollow first cavity to cause the movement of the movable piston between a first position in which the fuel and water is loaded into the hollow first cavity and a second position in which the residual fuel is moved from the hollow first cavity to the oxidation reactor (**312**).

(53) The techniques described herein have significant advantages. They facilitate the purification of

a contaminated liquid (e.g., water) using waste heat and energy from an oxidation reactor, where the reactor is fueled with the contaminants extracted from the contaminated liquid.

(54) A number of embodiments have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the invention.

(55) In addition, the logic flows depicted in the figures do not require the particular order shown, or sequential order, to achieve desirable results. In addition, other steps may be provided, or steps may be eliminated, from the described flows, and other components may be added to, or removed from, the described systems. Accordingly, other embodiments are within the scope of the following claims.

Claims

1. A system comprising: a supercritical water oxidation reactor configured to oxidize combustible fuel to produce heat and hot steam as a result of oxidation of the fuel; a first chamber having a hollow first cavity configured to contain water and the fuel; a vacuum pump operationally connected to the hollow first cavity and configured to reduce a pressure in the hollow first cavity to lower a boiling point of water in the hollow first cavity to a lowered boiling point below 192° F.; a second chamber including a hollow second cavity that at least partially surrounds the hollow first cavity, the hollow second cavity being operationally connected to the supercritical water oxidation reactor to receive the hot steam from the supercritical water oxidation reactor, such that hot steam received from the supercritical water oxidation reactor into the hollow second cavity causes water in the hollow first cavity to boil to cold steam; a movable piston located within the hollow first cavity and in sealable contact with an inner wall of the first chamber and configured to move along a length of the first chamber to mechanically move residual fuel in the hollow first cavity to the supercritical water oxidation reactor, wherein the residual fuel remains in the first cavity after water in the hollow first cavity boils to cold steam before the residual fuel is moved to the supercritical water oxidation reactor; and a hydraulic pump operationally connected to the hollow first cavity and configured to pump liquid in and out of the hollow first cavity to move the movable piston between a first position in which the fuel and water is in the hollow first cavity and a second position in which the residual fuel is moved from the hollow first cavity to the supercritical water oxidation reactor.
2. The system of claim 1, further comprising a heat exchanger configured for receiving the cold steam from the hollow first cavity and condensing the cold steam into a liquid.
3. The system of claim 1, further comprising a steam condensate tank operationally connected to the hollow second cavity and configured for receiving condensate produced in response to heat transferred from the hot steam in the hollow second cavity to the water and fuel in the hollow first cavity.
4. The system of claim 1, further comprising a drive system operationally coupled to the supercritical water oxidation reactor and to the hollow second cavity between the supercritical water oxidation reactor and the hollow second cavity and configured for receiving the hot steam from the supercritical water oxidation reactor and converting energy in the hot steam into mechanical energy and then passing the hot steam to the hollow second cavity.
5. The system of claim 1, wherein the supercritical water oxidation reactor is configured such that the oxidation of the fuel occurs at a temperature of greater than 500° F. to produce the heat and hot steam.
6. The system of claim 1, wherein the lowered boiling point is below 102° F.
7. The system of claim 1, wherein the hollow first cavity includes a first end operationally coupled to the hydraulic pump and a second end operationally coupled to the vacuum pump, the first end and the second end being separated by the movable piston in the hollow first cavity.
8. The system of claim 1 wherein the movable piston includes stainless steel and a sealing surface

on an outer surface of the piston, wherein the sealing surface seals against the inner wall of the first chamber, wherein the sealing surface is configured to remove condensate material from the inner wall of the first chamber.

9. The system of claim 1, wherein the first chamber includes at least one copper wall defining the hollow first cavity.

10. The system of claim 1, wherein the second chamber includes at least one copper wall defining the hollow second cavity.

11. A method comprising: oxidizing combustible fuel in a supercritical water oxidation reactor to produce heat and hot steam as a result of oxidation of the fuel; loading a hollow first cavity of a first chamber that is operationally connected to the supercritical water oxidation reactor with water and the fuel; reducing a pressure in the hollow first cavity to lower a boiling point of water in the hollow first cavity to a lowered boiling point below 192° F.; providing the hot steam from the supercritical water oxidation reactor to a hollow second cavity of a second chamber that at least partially surrounds the hollow first cavity, such that hot steam received from the supercritical water oxidation reactor into the hollow second cavity causes water in the hollow first cavity to boil to cold steam at the lowered boiling point; moving a movable piston located within the hollow first cavity and that is in sealable contact with an inner wall of the first chamber along a length of the first chamber to mechanically move residual fuel in the hollow first cavity to the supercritical water oxidation reactor, wherein the residual fuel remains in the hollow first cavity after water in the hollow first cavity boils to cold steam before the residual fuel is moved to the supercritical water oxidation reactor, wherein residual fuel that is moved into the supercritical water oxidation reactor provides the combustible fuel for oxidation in the supercritical water oxidation reactor; and pumping liquid in and out of the hollow first cavity to cause movement of the movable piston between a first position in which the fuel and water is loaded into the hollow first cavity and a second position in which the residual fuel is moved from the hollow first cavity to the supercritical water oxidation reactor.

12. The method of claim 11, further comprising condensing the steam from the hollow first cavity into a liquid.

13. The method of claim 11, further comprising receiving into a steam condensate tank that is operationally connected to the hollow second cavity condensate produced in response to heat transferred from the hot steam in the hollow second cavity to the of water and fuel in the hollow first cavity.

14. The method of claim 11, further comprising: receiving into a drive system operationally coupled to the supercritical water oxidation reactor and to the hollow second cavity between the supercritical water oxidation reactor and the hollow second cavity the hot steam from the supercritical water oxidation reactor; converting energy in the hot steam into mechanical energy; and passing the hot steam to the hollow second cavity.

15. The method of claim 11, wherein the oxidation of the fuel in the supercritical water oxidation reactor occurs at a temperature of greater than 500° F.

16. The method of claim 11, wherein the lowered boiling point is below 102° F.

17. The method of claim 11, wherein the hollow first cavity includes a first end operationally coupled to a hydraulic pump that pumps the liquid in and out of the hollow first cavity and a second end operationally coupled to the vacuum pump, the first end and the second end being separated by the movable piston in the hollow first cavity.

18. The method of claim 11, wherein the movable piston includes stainless steel and a sealing surface on an outer surface of the piston, wherein the sealing surface seals against the inner wall of the first chamber, wherein the sealing surface is configured to remove condensate material from the inner wall of the first chamber.

19. The method of claim 11, wherein the first chamber includes at least one copper wall defining the hollow first cavity.

20. The method of claim 11, wherein the second chamber includes at least one copper wall defining the hollow second cavity.
